

Applied Mathematics and Informatics In Drug Discovery (2026)

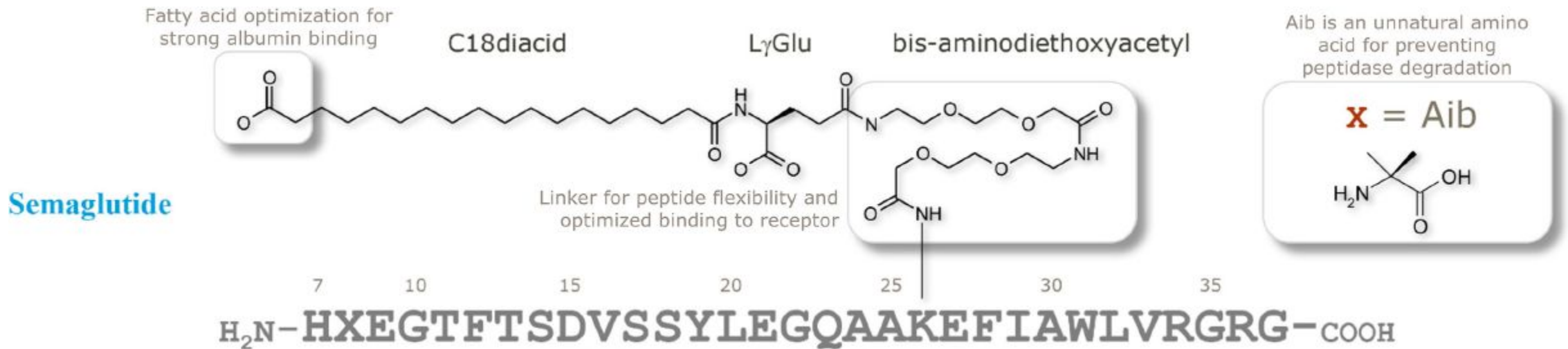


Figure from [Knudsen et al., 2019](#)

PD Dr. Jitao David Zhang, Computational Biologist

¹ Computational Sciences Center of Excellence (CS CoE), Roche Innovation Center Basel, F. Hoffmann-La Roche;

² Department of Mathematics and Computer Sciences, University of Basel

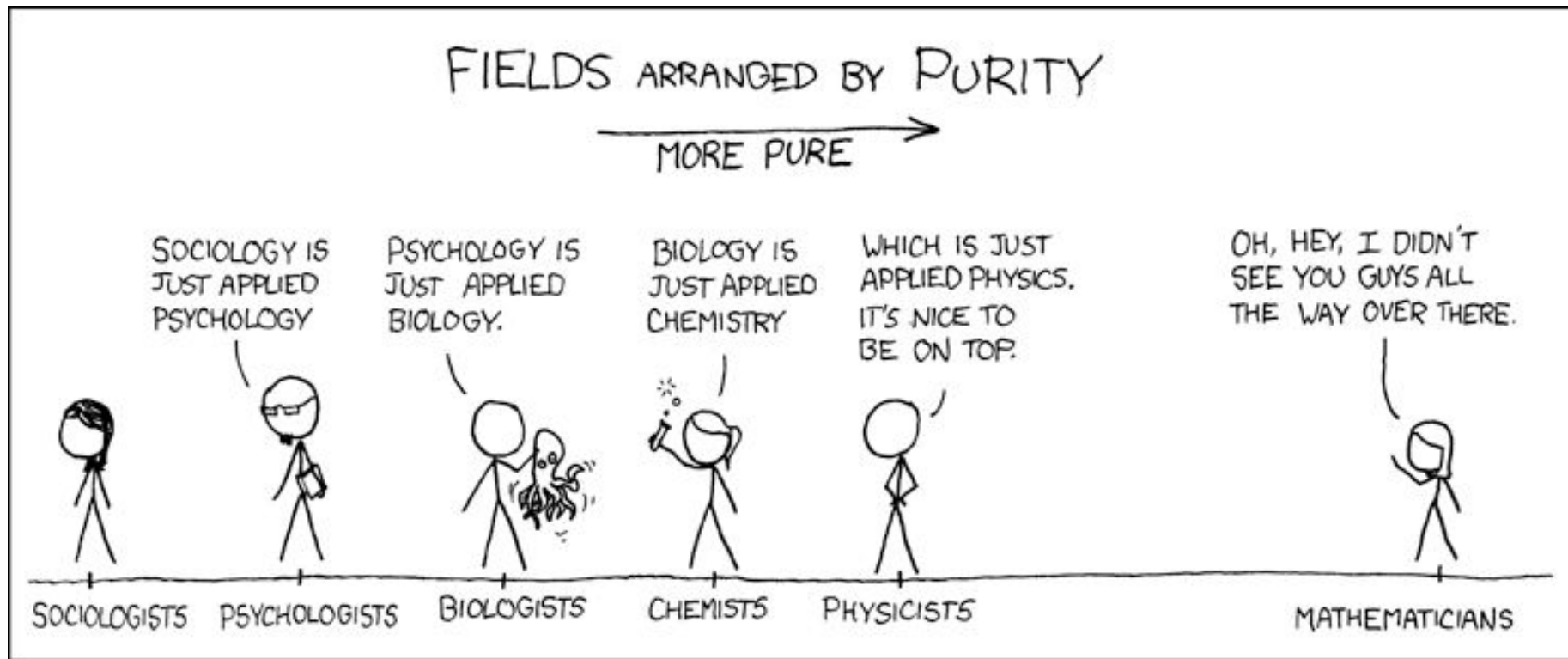
Outline

1. About the course
2. Drug discovery and development: The example of semaglutide
3. The multiscale view of drug discovery: A process to understand biology and consequences of body-drug interactions

About the course

Purity

<https://xkcd.com/435/>

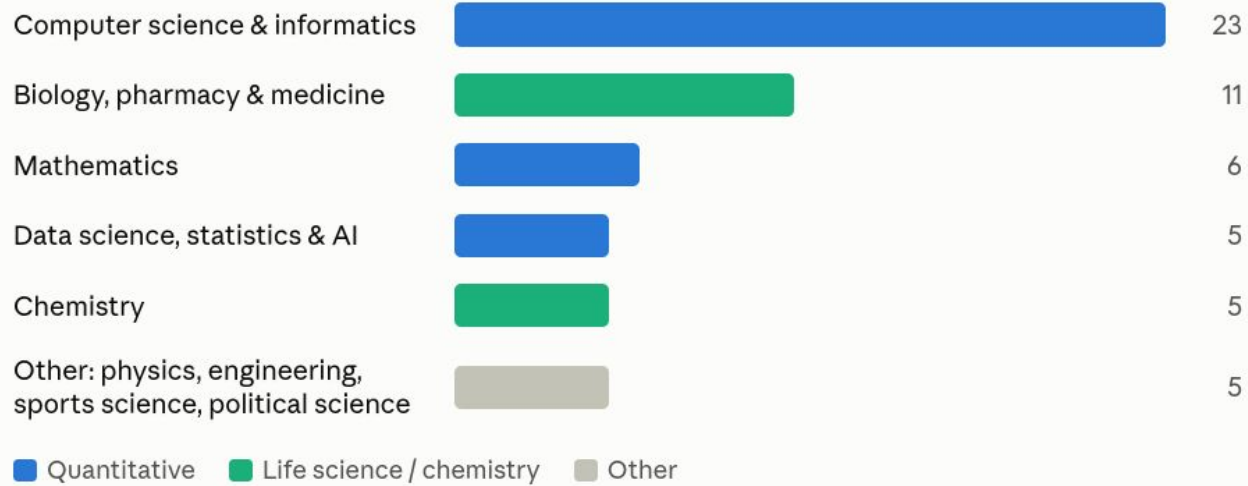


This course aims to bring people together and to promote interdisciplinary research

Our strength: we have a diverse classroom!

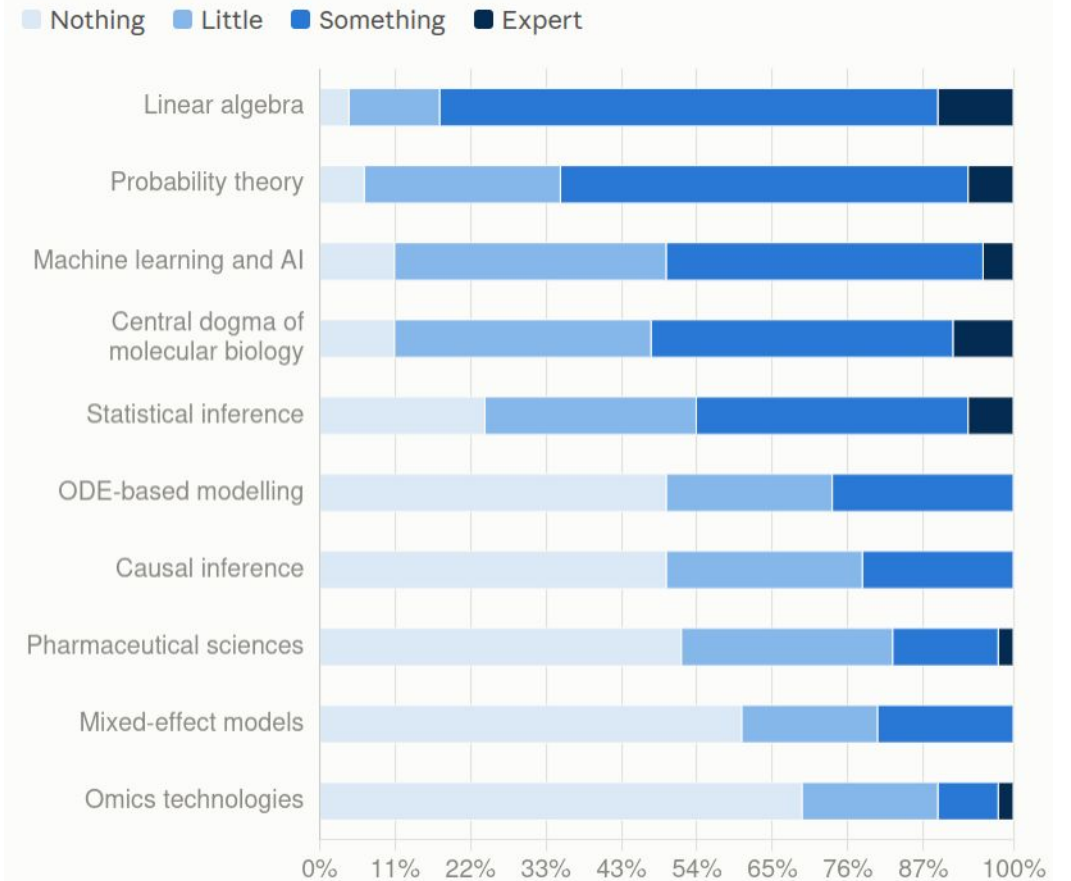
Background

Disciplines named (multiple answers allowed, n = 46)



Prior knowledge

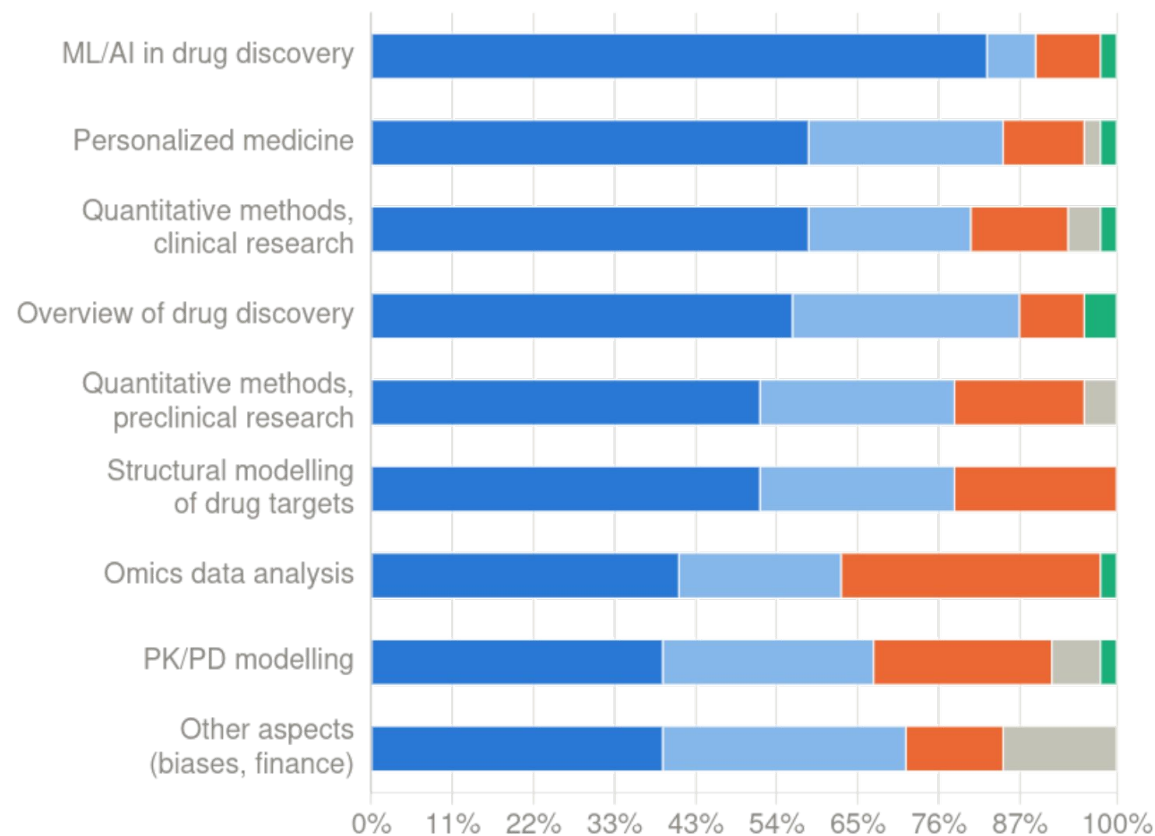
Self-rated prior knowledge, share of respondents



Where we want to go - together

Interest in course topics (n = 46)

■ Very interested
 ■ Somehow interested
 ■ Don't know it, tell me more
■ Not at all interested
 ■ Know it already



Representative motivations

1. **Applying mathematics and informatics to real and practical problems:** *"The chance to learn to apply my technical knowledge ... to something practical"*
2. **Curiosity about drug discovery, perhaps with career perspectives in mind:** *"I enjoy organic chemistry and could imagine pursuing a career in pharma. I believe that AI ... will/ already play a big role in the discovery of new compounds.", "to get job in pharma companies"*
3. **Bridging two worlds:** *"As a registered pharmacist pursuing Pharma Informatics, my goal is to serve as an integrator who seamlessly bridges clinical expertise and AI technology."*
4. **Interest in how AI will change drug discovery:** *"I want to take the AMIDD course because I am fascinated by how AI can speed up drug discovery and ultimately save patient lives."*
5. **Mandatory/required course**

Use of AI - let's make a contract

Profile #1: Enthusiastic about AI, not without cautions, though - especially notable among students with data-science background:

- "We should learn how to use them responsibly, without completely letting it take over our own thinking process."
- "discuss hallucinations, incorrect assumptions, reproducibility, and how to distinguish a plausible-looking answer from a scientifically valid one."

Profile #2: AI as sparring partner, tutor and explainer and leveller

- "Maybe we can use AI to create questions to think about or questions that deepen our understanding."
- "I mainly use LLM's as a tool to deepen my understanding of the material after lectures."
- "Maybe to breakdown complex literature into absorbable information."

Profile #3: AI as an assistant for data collection, coding, and analysis:

- "Through practical coding labs (as Jupyter Notebooks) and real case studies showing the success and limits of AI in the field"
- "Help us to write code to test stuff." / "Maybe to help analyze the datasets?" / "maybe to generate some visualisations for some processes"
- "I don't have precise suggestions, but I have a Claude Pro subscription and use it daily. I am very interested to learn how to apply it more effectively to work in this field."

Profile #4: Wanting to learn how others use AI, especially those in the industry

- "I'm not sure, because I have never used AI for research, but I would like to learn how AI is used in a research setting, just like experts would use it."
- "We can learn about the artificial intelligence tools used in the industry."
- "I would be glad to provide this feedback once I understand what is currently being employed and the policies around use of AI for such disciplines."

Profile #5: Reserved or undecided

roughly a quarter gave a non-answer ("I have no idea", "No idea", "I don't know.", "-"), plus one outright minimalist: "best would be as little as possible." Two deferred it back to you: "Tough question, let's discuss in class?" and "I will probably have an answer to this at the end of the course."

Course information for AMIDD 2026

- Lecturer: Jitao David Zhang
(jitao-david.zhang@unibas.ch)
- Website: [AMIDD.ch](http://www.amidd.ch)
- Thirteen lectures this semester
 - Introduction (1 session)
 - Mechanistic, statistical, and causal models (2 sessions)
 - Molecular level modelling (2 sessions)
 - Omics- & cellular models (2 sessions)
 - Organ- and system models (2 sessions)
 - Population modelling (2 sessions)
 - Invited guest speakers (1 session)
 - A collaboration challenge (1 session)
- Fridays 12:15-14:00
- Slides, exercises, pre-reading and post-reading articles are shared on the course's website <http://www.amidd.ch>. We do not provide recordings.
- The final grade is given by participation including quizzes (30%), offline activities (40%), and a collaboration challenge in the final session (30%). The topic of collaboration challenge will be announced in the second to last session.
- **Questions?**

I look forward to sharing my experience in drug discovery and learning from you!

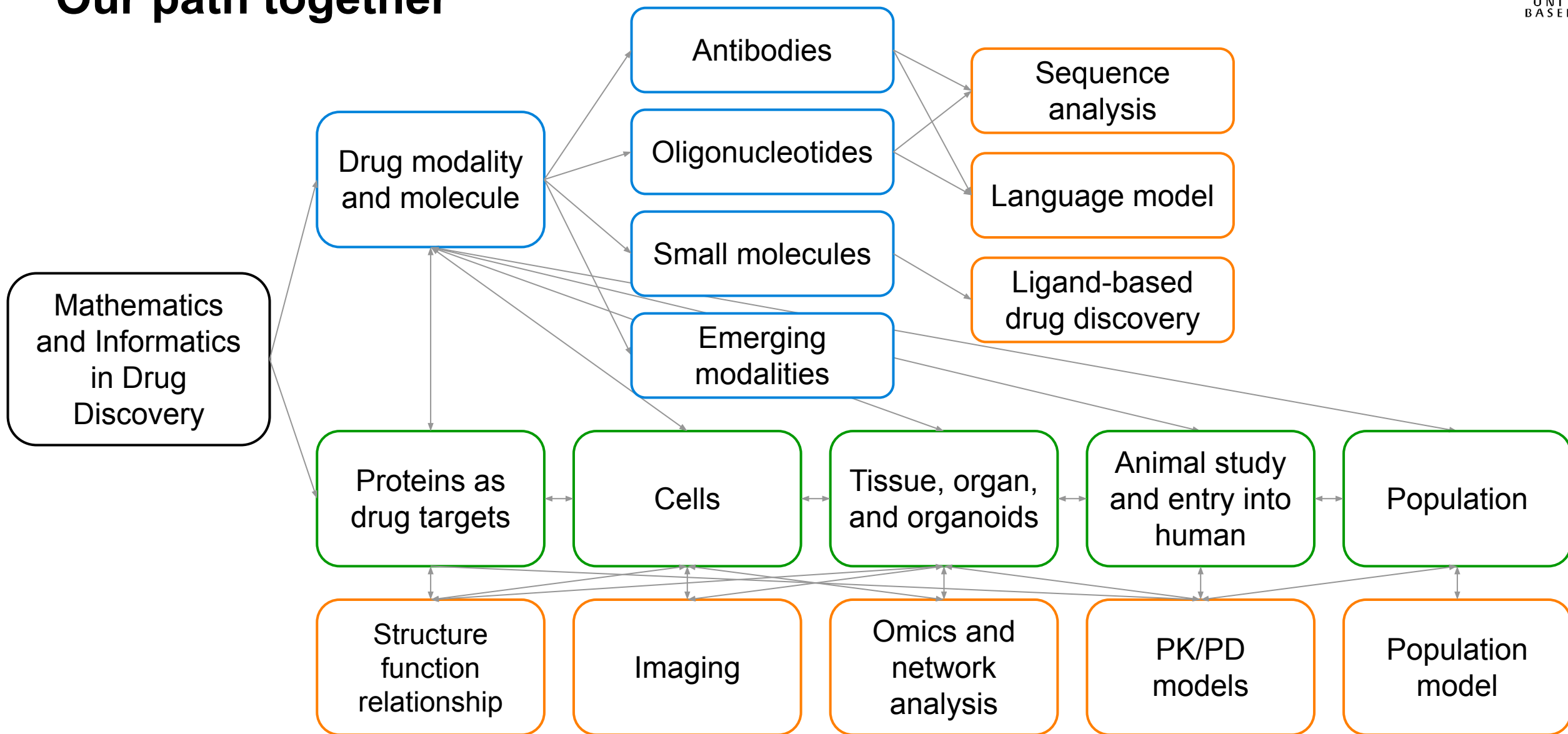
Disclaimer

Teaching is my personal engagement. My opinions and views do not necessarily reflect those of F. Hoffmann-La Roche, my employer.

Please be aware of my biases and limitations.

- I am a computational biologist working in drug discovery, with limited understanding of mathematics, computer science, biology, chemistry, pharmacology, toxicology, and medicine.
- I see my task is as sharing with you the mathematical concepts and computational approaches used in drug discovery that I find beautiful and useful.
- I look forward to learning from you mathematics and other expertise from you.

Our path together



Drug discovery and development: The example of semaglutide

Oral Semaglutide in Adults with Overweight or Obesity



For participants with overweight or obesity, the efficacy and safety of oral semaglutide at a dose of 25 mg once daily for weight management are unclear. New research findings are summarized in a short video, available at [NEJM](#) (published on September 18, 2025)

Q1: What is semaglutide?

Q2: What are the proven benefits of semaglutide for participants with obesity?

Q3: What is the dose, the dosing frequency, and the route of administration that has been shown to provide a benefit?

Q4: What is the goal of this study?

Q5: How many participants participated in the trial?

Q6: Which participants participated in the trial?

Q7: What were the coprimary outcomes of the trial?

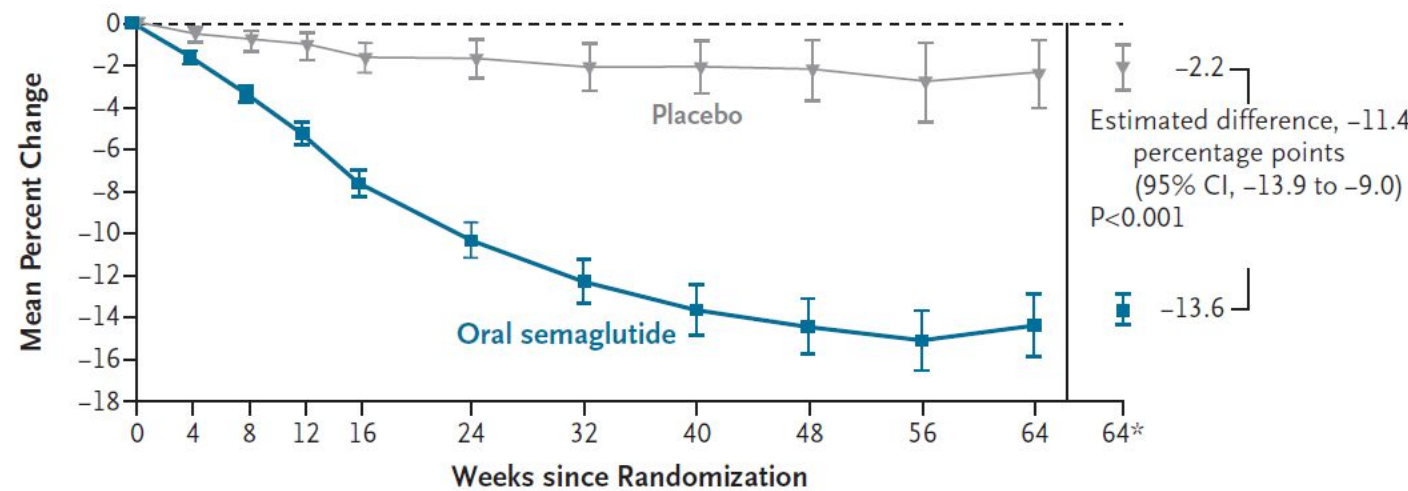
Q8: What was the most common adverse event? How many participants experienced that?

Q9: What was the conclusion of the authors?

Q10: What questions do you have?

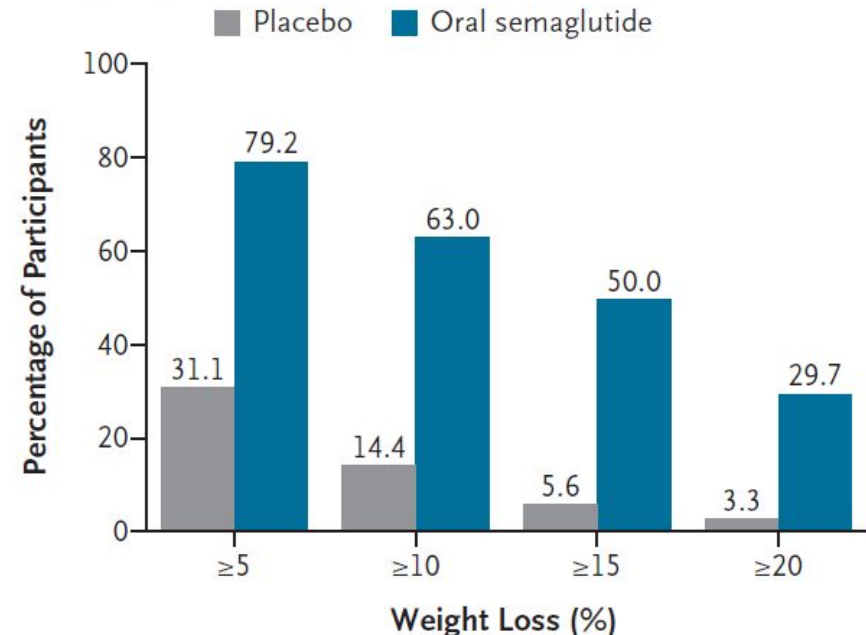
Efficacy and Safety of Oral Semaglutide 25 mg Once Daily in Adults With Overweight or Obesity ([The OASIS 4 study](#))

A Change in Body Weight from Baseline



No. of Participants												
Placebo	102	100	99	99	99	92	89	89	88	87	90	102
Oral semaglutide	205	201	200	200	198	192	189	181	186	179	192	205

B Participants Meeting Weight-Loss Targets at Week 64



Panel A shows mean percent change in body weight from baseline over time. Error bars indicate 95% confidence intervals (CIs), numbers below show participants, and asterisks mark estimated means. Panel B shows percentages of participants with ≥5%, ≥10%, ≥15%, and ≥20% weight loss at week 64.

Primary endpoints (not complete)

Table 2. Primary, Confirmatory Secondary, and Selected Exploratory End Points.*

End Point	Oral Semaglutide (N = 205)	Placebo (N = 102)	Difference (95% CI)†
Primary end points			
Percent change in body weight from baseline to week 64	−13.6	−2.2	−11.4 (−13.9 to −9.0)
Body-weight reduction of ≥5% at week 64 — no./total no. (%)‡	152/192 (79.2)	28/90 (31.1)	7.3 (4.2 to 12.8)
Confirmatory secondary end points			
Body-weight reduction of each target — no./total no. (%)‡			
≥10% at week 64	121/192 (63.0)	13/90 (14.4)	9.1 (4.7 to 17.3)
≥15% at week 64	96/192 (50.0)	5/90 (5.6)	15.7 (6.2 to 40.2)
≥20% at week 64	57/192 (29.7)	3/90 (3.3)	12.2 (3.7 to 40.3)
Change in IWQOL-Lite-CT Physical Function score from baseline to week 64 — points§	16.2	8.4	7.7 (3.3 to 12.2)
Supportive secondary end points¶			
Change in IWQOL-Lite-CT Physical Function score ≥14.6 at week 64 — no./total no. (%)‡	104/188 (55.3)	31/89 (34.8)	2.4 (1.4 to 4.1)
Change in cardiometabolic risk factors from baseline to week 64			
Body weight — kg	−14.2	−2.16	−12.0 (−14.6 to −9.5)
BMI	−5.1	−0.8	−4.3 (−5.2 to −3.4)

CI: confidence interval

IWQOL: A five-item subscore of the Impact of Weight on Quality of Life–Lite Clinical Trials Version (IWQOL-Lite-CT), in which scores range from 0 to 100, with higher scores indicating better level of function. It considers physical function, self-esteem, sexual life, public distress, and work.

BMI: Body-mass Index, the weight in kilograms divided by the square of the height in meters

Safety outcome, i.e. Adverse events

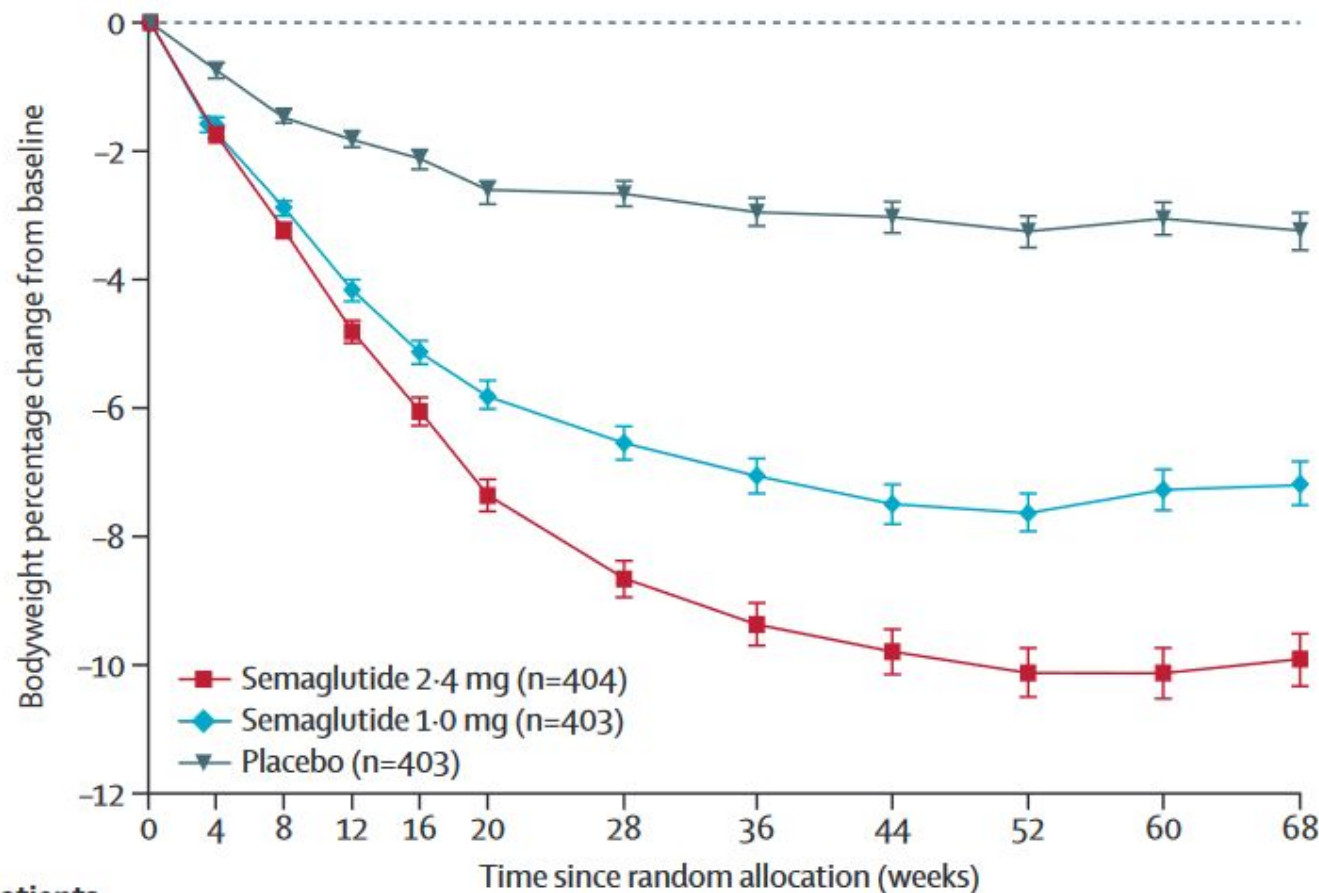
Included are all adverse events that occurred in the safety analysis population, which include all the participants who underwent randomization and were exposed to at least one dose of semaglutide or placebo.

Adverse events were classified by severity as *mild* (causing minimal discomfort and not interfering with everyday activities), *moderate* (causing sufficient discomfort to interfere with normal everyday activities), or *severe* (preventing normal everyday activities).

Table 3. Adverse Events.*

Event	Oral Semaglutide (N = 204)			Placebo (N = 102)		
	no. of participants (%)	no. of events	events per 100 participant-yr of exposure	no. of participants (%)	no. of events	events per 100 participant-yr of exposure
Any adverse event	190 (93.1)	1239	493.5	87 (85.3)	432	355.9
Serious adverse events	8 (3.9)	17	6.8	9 (8.8)	13	10.7
Adverse events leading to discontinuation of semaglutide or placebo	14 (6.9)	14	5.6	6 (5.9)	6	4.9
Gastrointestinal disorders	7 (3.4)	7	2.8	2 (2.0)	2	1.6
Fatal events	0	0	0	0	0	0
Adverse events reported in ≥10% of participants						
Nausea	95 (46.6)	157	62.5	19 (18.6)	27	22.2
Vomiting	63 (30.9)	105	41.8	6 (5.9)	6	4.9
Nasopharyngitis	43 (21.1)	59	23.5	27 (26.5)	40	33.0
Coronavirus disease 2019	42 (20.6)	46	18.3	18 (17.6)	19	15.7
Constipation	41 (20.1)	59	23.5	10 (9.8)	11	9.1
Dyspepsia	37 (18.1)	50	19.9	9 (8.8)	11	9.1
Diarrhea	36 (17.6)	61	24.3	9 (8.8)	10	8.2
Headache	24 (11.8)	35	13.9	9 (8.8)	10	8.2
Eructation	21 (10.3)	23	9.2	2 (2.0)	2	1.6

STEP 2: a clinical trial for Semaglutide 2.4 mg once a week for overweight, obesity, and type-2 diabetes

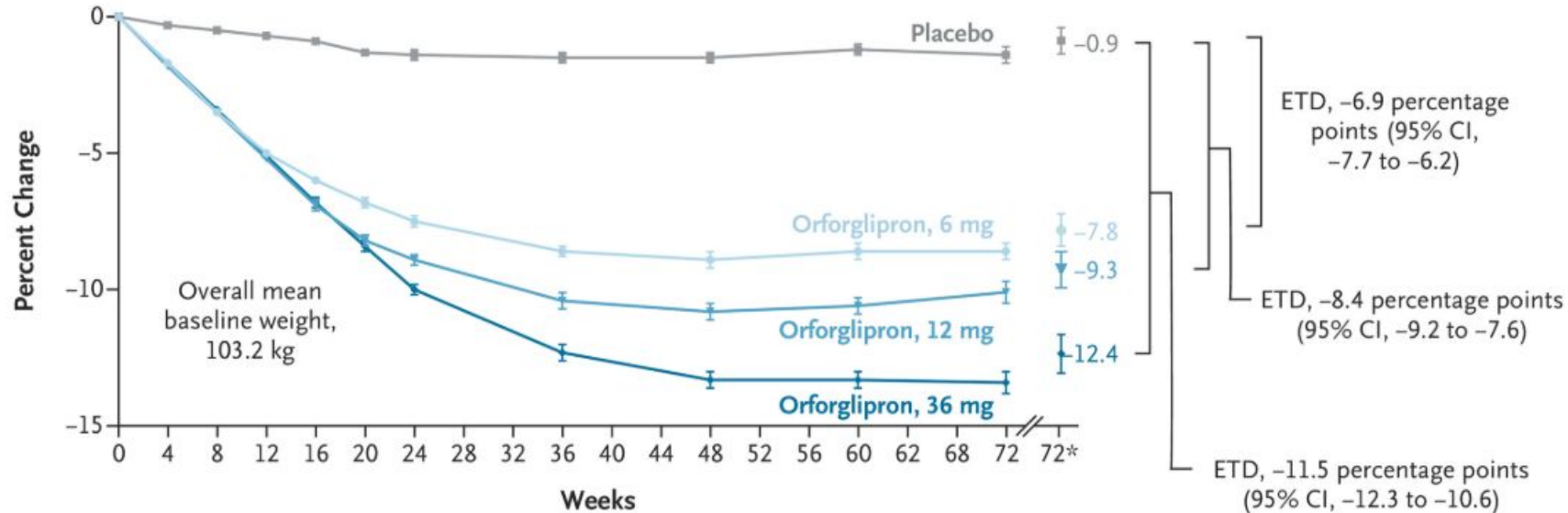


[Semaglutide 2.4 mg once a week in adults with overweight or obesity, and type 2 diabetes \(STEP 2\): a randomised, double-blind, double-dummy, placebo-controlled, phase 3 trial](#), Davies *et al.*, 2021

Number of patients												
Semaglutide 2.4 mg	404	395	397	390	388	392	386	383	381	381	378	388
Semaglutide 1.0 mg	403	394	392	385	383	383	378	377	373	370	374	380
Placebo	403	398	394	389	387	383	381	377	371	367	366	376

ATTAIN-1: A Study of Orforglipron (LY3502970) in Adult Participants With Obesity or Overweight With Weight-Related Comorbidities

B Change in Body Weight from Baseline to Week 72 (efficacy estimand)



No. of Patients

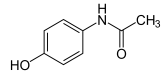
Placebo	949	937	923	910	875	847	817	766	716	688	654	949
Orforglipron, 6 mg	723	713	702	690	675	664	652	630	601	578	559	723
Orforglipron, 12 mg	725	719	695	687	672	652	648	624	602	582	559	725
Orforglipron, 36 mg	730	715	704	689	670	662	644	617	583	573	549	730

[Orforglipron, an Oral Small-Molecule GLP-1 Receptor Agonist for Obesity Treatment](#), Wharton *et al.*, 2025.

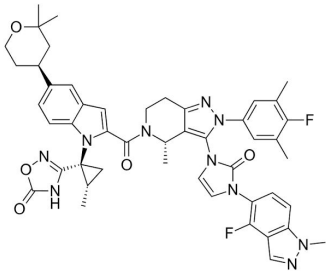
ETD: estimated treatment difference.

Orforglipron (*Foundayo*) is an oral GLP-1 receptor agonist developed as a weight loss drug by Eli Lilly and Company. It was discovered by Chugai Pharmaceutical Co. and licensed to Lilly in 2018. It was approved for medical use in the U.S. in April 2026, and in the UK in August 2026.

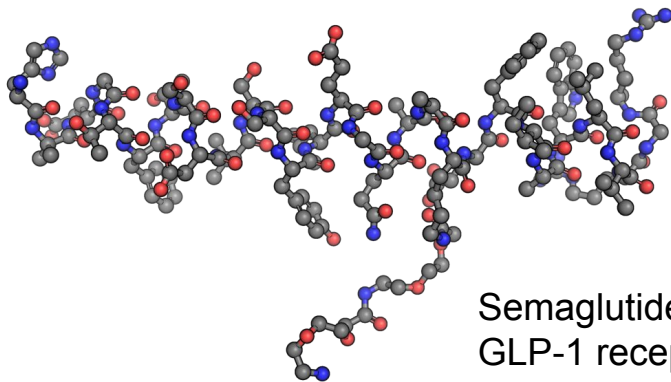
Small molecule, peptide, and antibody



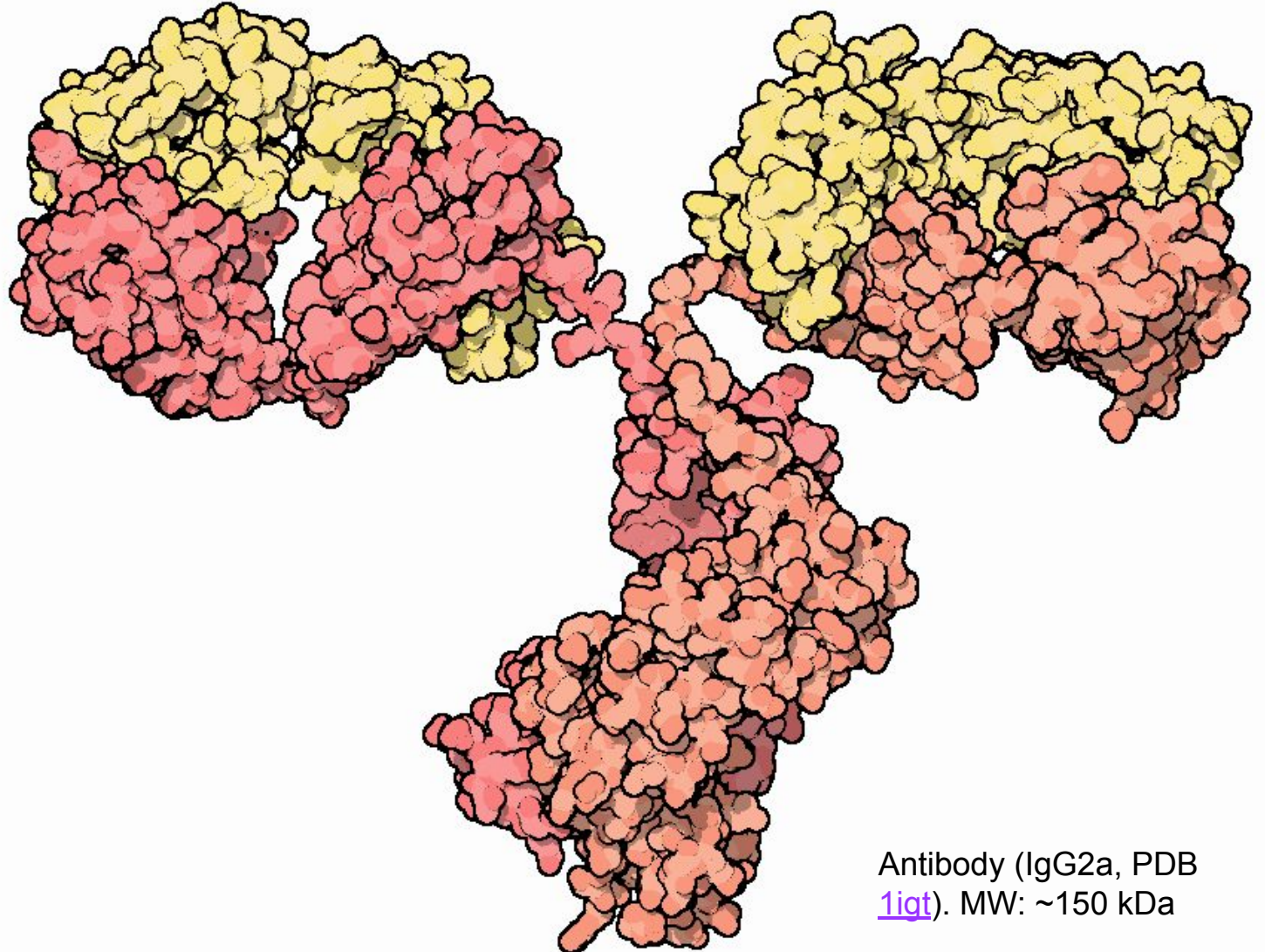
Paracetamol (oral, non-opioid analgesic and antipyretic). Molecular Weight (MW): 151 Dalton (g/mol)



Orforglipron (oral, non-peptide, small molecule GLP-1 receptor agonist). MW: 883 Dalton.

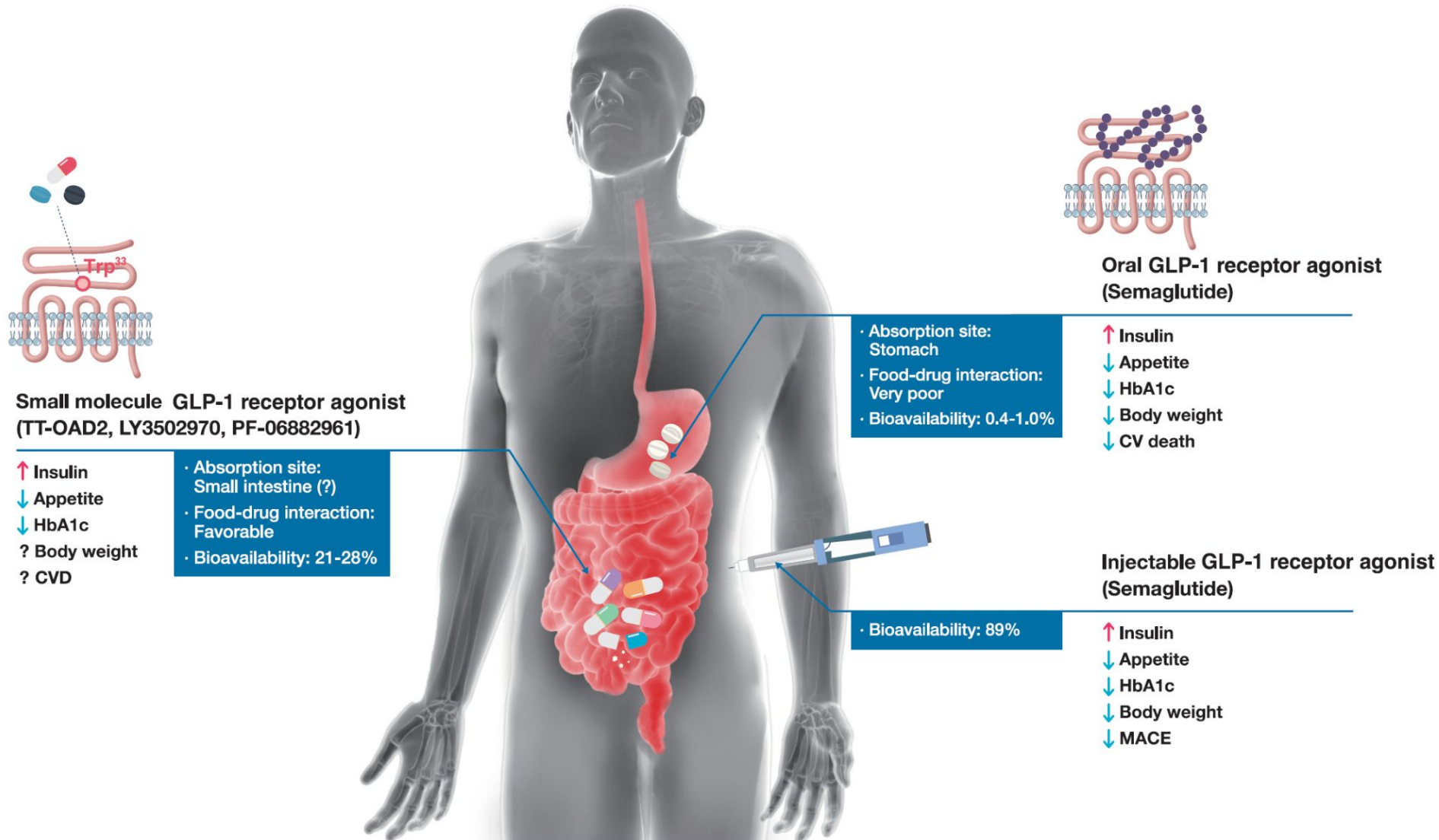


Semaglutide (peptide GLP-1 receptor agonist). MW: ~4000 Dalton (~4 kDa)



Antibody (IgG2a, PDB [1igt](#)). MW: ~150 kDa

Comparing injectable semaglutide, oral semaglutide, and small-molecule GLP1R agonist



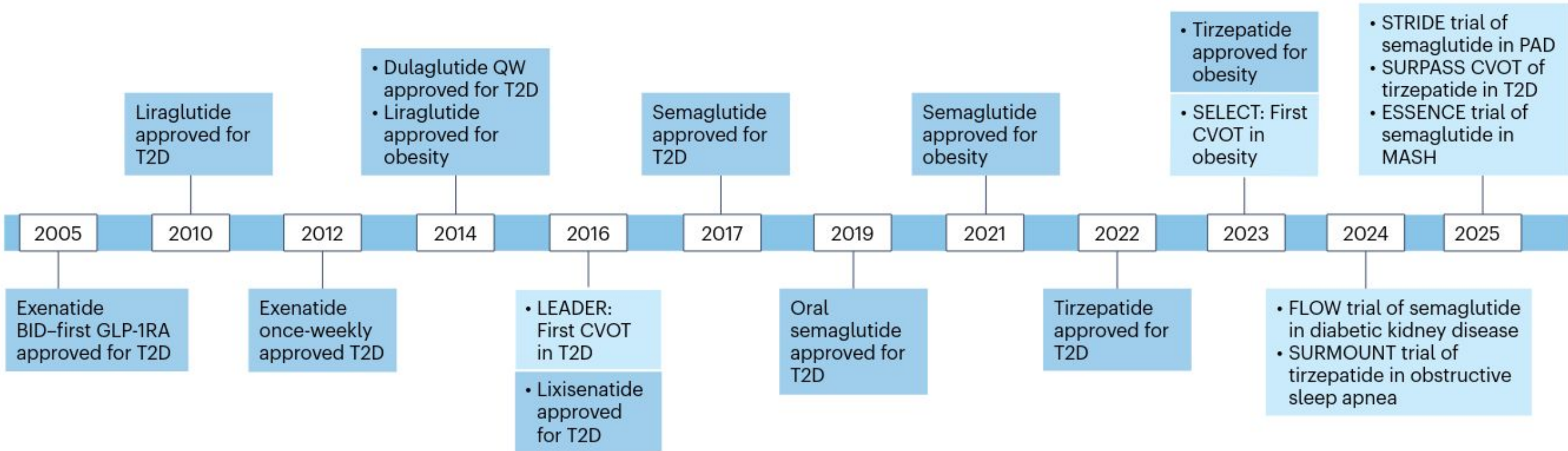
HbA1c: Hemoglobin A1c, hemoglobin chemically linked to a sugar. A HbA1c test shows your average level of blood glucose, i.e. blood sugar, over the past two to three months. HbA1c is therefore a *biomarker* of blood sugar.

CVD: Cardiovascular disease.

MACE: Major adverse cardiovascular event.

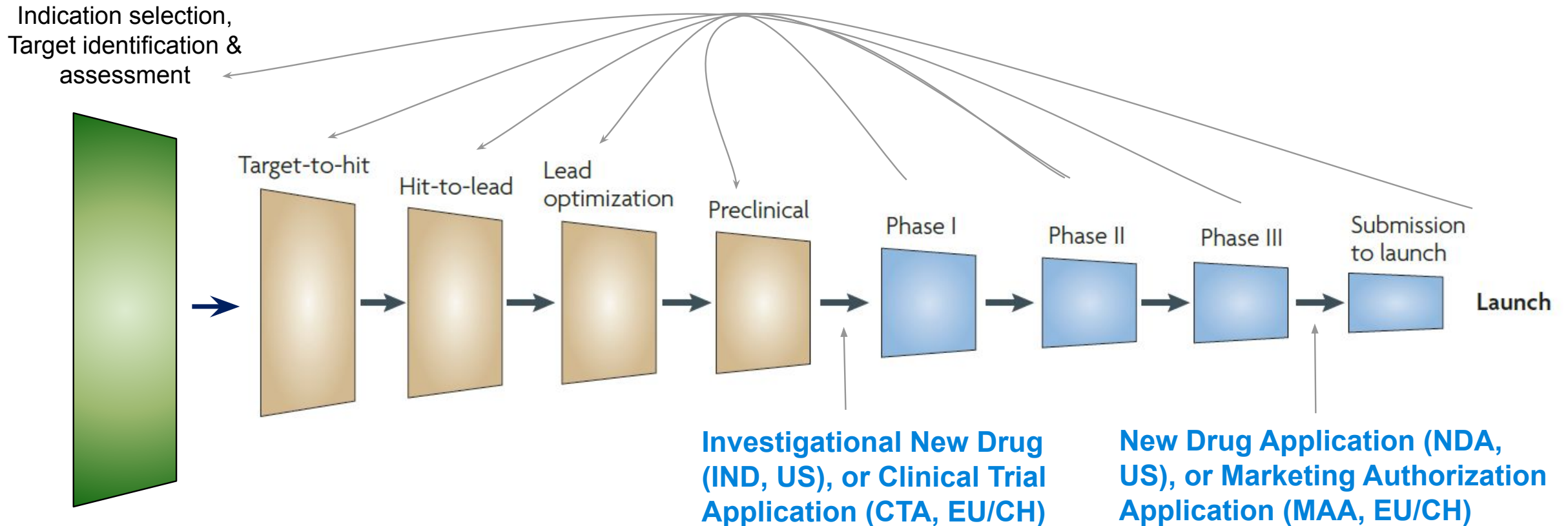
Choe, Hun Jee, and Young Min Cho. 2021. "Peptidyl and Non-Peptidyl Oral Glucagon-Like Peptide-1 Receptor Agonists." *Endocrinology and Metabolism* 36 (1): 22–29. <https://doi.org/10.3803/EnM.2021.102>.

Timeline of major trials and approvals for GLP-1 medicines



Key approval dates for individual medicines and indications are indicated with dark-blue shading. Key outcome trials are indicated in light blue. BID, twice daily; CVOT, cardiovascular outcomes trial; MASH, metabolic dysfunction-associated steatohepatitis; PAD, peripheral artery disease.

The linear view of drug discovery



Adapted from Paul *et al.* "How to Improve R&D Productivity: The Pharmaceutical Industry's Grand Challenge." *Nature Reviews Drug Discovery*, 2010

The multiscale view of drug discovery

Glucagon-like peptide 1 (GLP-1) and chemical derivatives

GLP-1 is a 30- or 31-amino acid long peptide. It is derived from a protein (preproglucagon) encoded by the gene GCG in human.

Endogenous GLP-1 has a half-life of about 2 minutes, making the natural form a poor drug candidate due to its pharmacokinetic (PK) behaviour.

Synthesized agonists have similar sequences but much longer half-life due to chemical modifications. Examples include:

- Semaglutide (top, Novo Nordisk, approved, half-life: about 1 week)
- Taspoglutide (right middle panel, tested in Phase III trials, discontinuation due to hypersensitivity and GI tolerability)

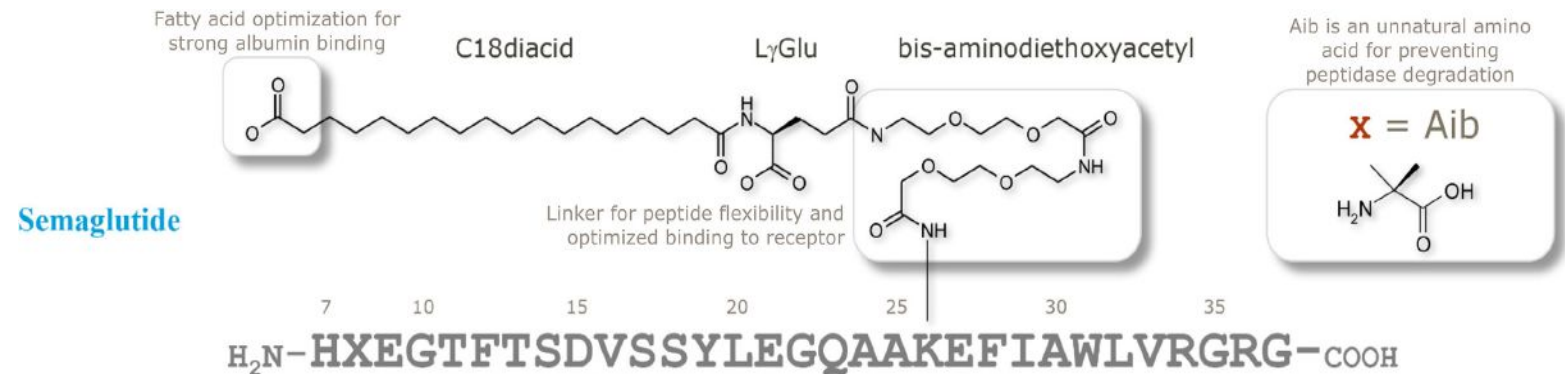


Figure from [Knudsen et al., 2019](#)

SEMAGLUTIDE	1 HXEGTFTSDVSSYLEGQAAKEFIAWLVRGRG	31
GLP-1	1 HAEGTFTSDVSSYLEGQAAKEFIAWLKGRG	31
TASPOGLUTIDE	1 HXEGTFTSDVSSYLEGQAAKEFIAWLKXR	30
GLP-1	1 HAEGTFTSDVSSYLEGQAAKEFIAWLKGR	30

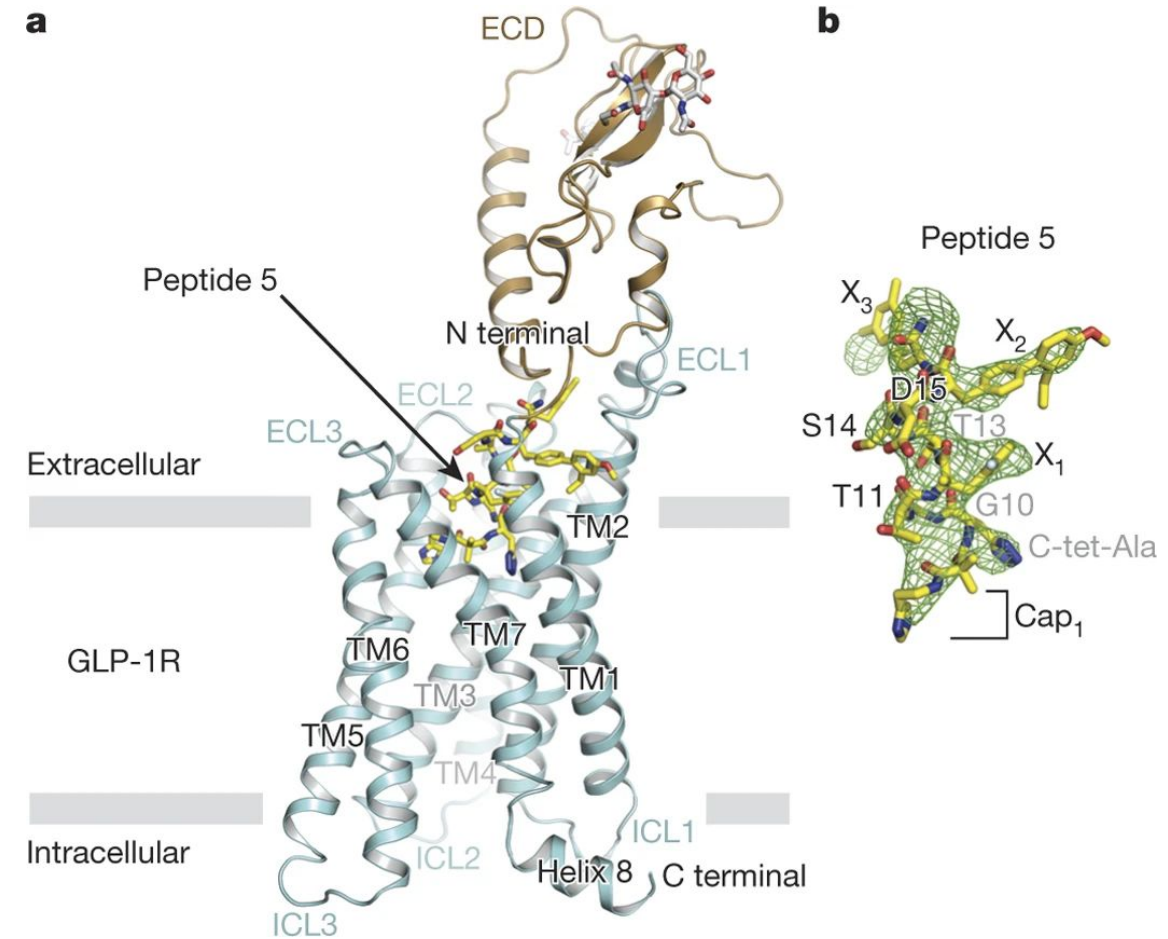
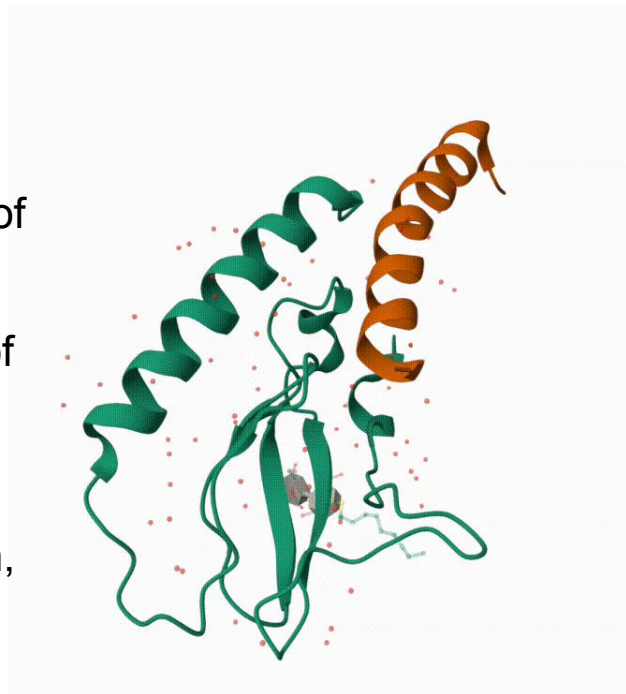
GLP-1 binds to its receptor GLP1R to induce insulin secretion

GLP-1 is secreted from the intestine after nutrient intake. GLP-1 binds to its receptor GLP1R, and the binding event activates intracellular signaling and stimulates glucose-dependent insulin secretion.

GLP1R is a GPCR (G-protein coupled receptor) found on the surface of cells in pancreas, and other cells in the body.

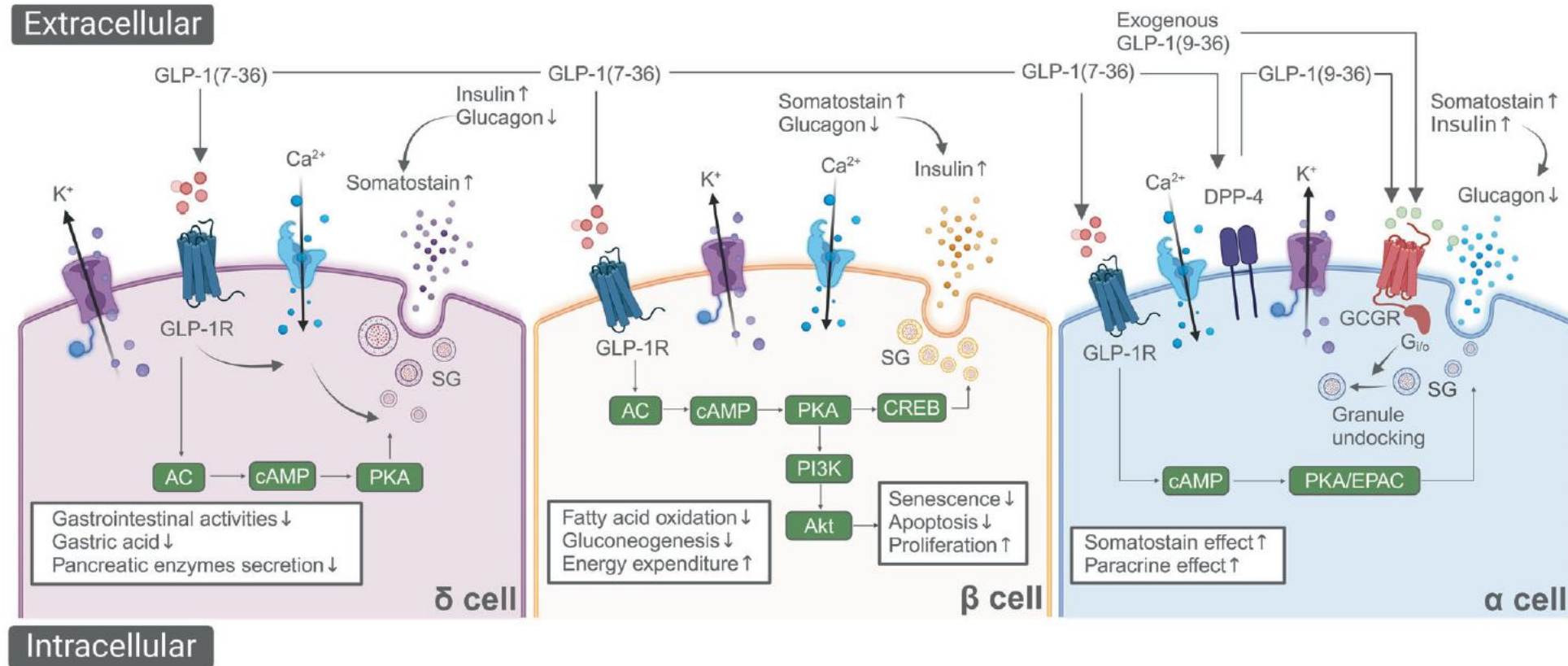
Left: crystal structure of GLP-1 (orange) bound to a fragment of GLP1R (green) (PDB: [3IOL](#))

Right: Ribbon representation of GLP1R bound to an agonist peptide. The transmembrane domain, or TMD, is coloured cyan. The extracellular domain, or ECD, is coloured brown.



A. Jazayeri et al. (2017), "Crystal structure of the GLP-1 receptor bound to a peptide agonist," <https://doi.org/10.1038/nature22800>, Nature

Mechanisms of Blood Glucose Reduction by GLP-1 in Pancreatic α (alpha), β (beta), and δ (delta) Cells



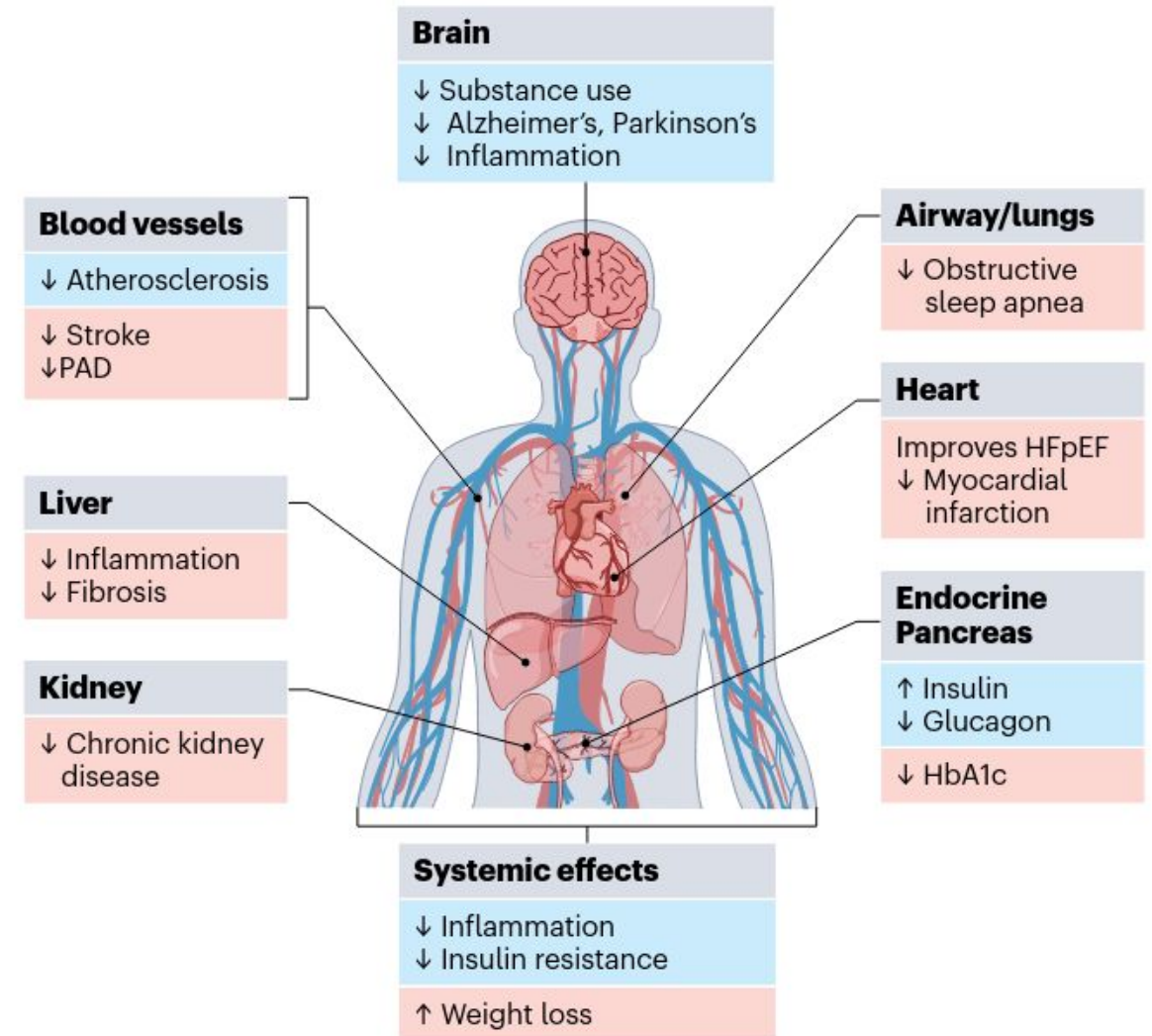
The cartoons give a simplified summary of our current understanding on how GLP-1 reduces blood glucose levels by acting on different pancreatic cell types. Direct and indirect effects of GLP-1 on other cells may employ other mechanisms.

Organ-level effects and assumed benefits of action of GLP-1 medicine

The benefits of GLP-1 medicines have been demonstrated in multiple outcome trials for people with

- cardiovascular disease,
- peripheral artery disease (PAD),
- liver disease, kidney disease,
- Heart Failure with preserved Ejection Fraction (HFpEF), and
- obstructive sleep apnea

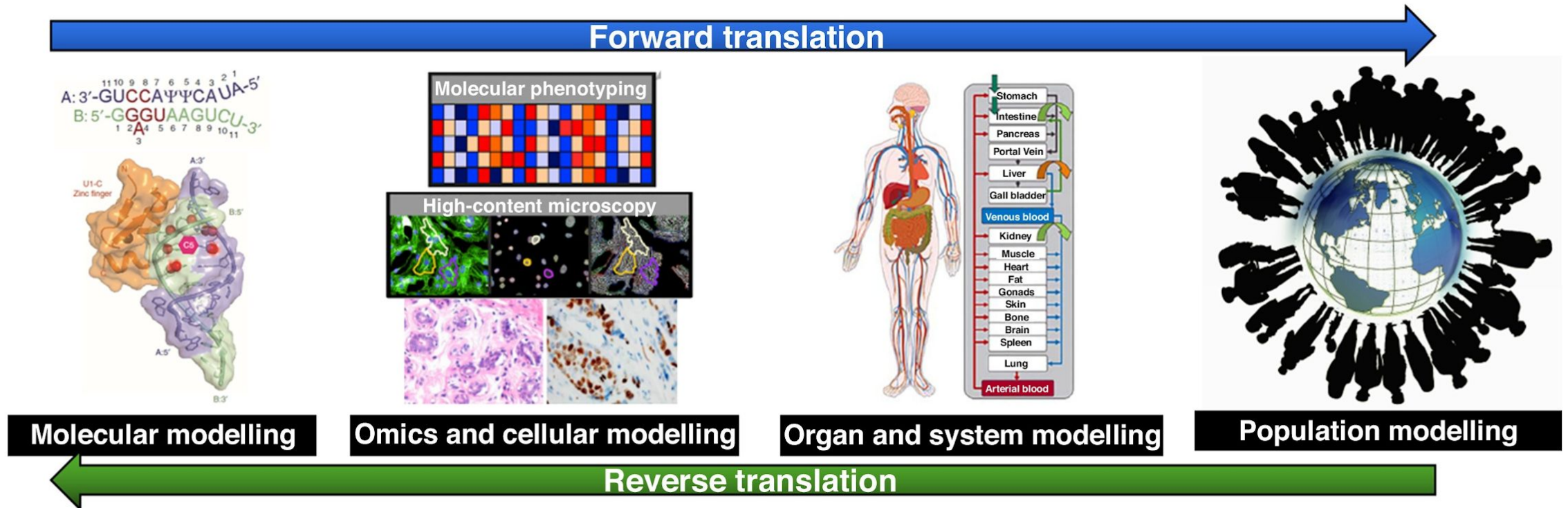
Multiple new GLP-1 medicines are under investigation for a wide range of indications.



Drucker, Daniel J. "The Expanding Landscape of GLP-1 Medicines." *Nature Medicine* 32, no. 1 (2026): 47–57. <https://doi.org/10.1038/s41591-025-04124-5>.

HbA1c: Hemoglobin A1c. Oral semaglutide Alzheimer's trials (EVOKE/EVOKE+) did not meet their primary endpoints by the end of 2025.

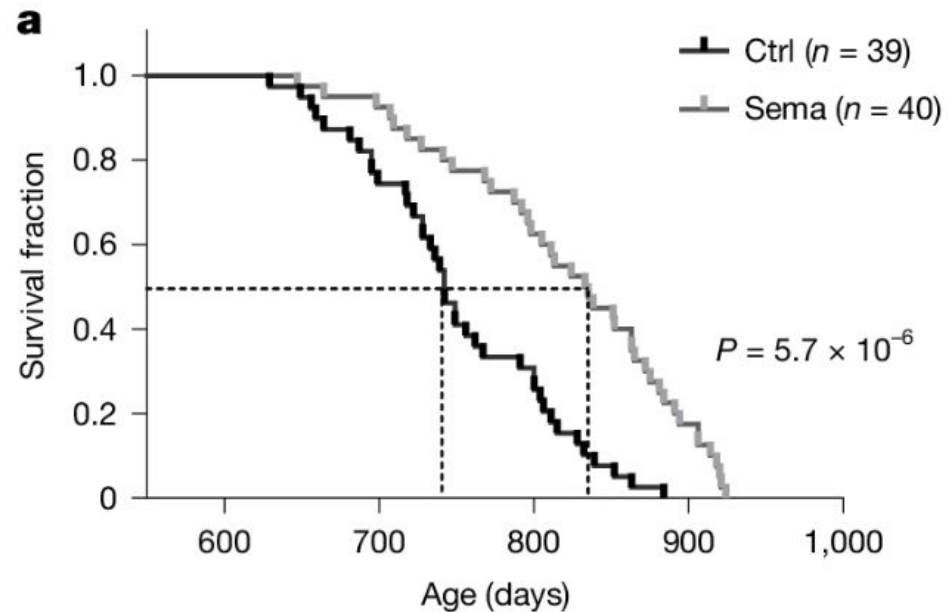
The multiscale modelling view of drug discovery



Drug Discovery Today

Zhang, Jitao David, Lisa Sach-Peltason, Christian Kramer, Ken Wang, and Martin Ebeling. 2020. "Multiscale Modelling of Drug Mechanism and Safety." *Drug Discovery Today* 25 (3): 519–34. <https://doi.org/10.1016/j.drudis.2019.12.009>.

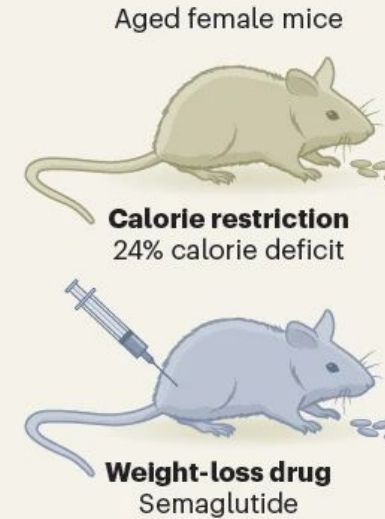
Late-life semaglutide treatment slows ageing and extends lifespan in female mice



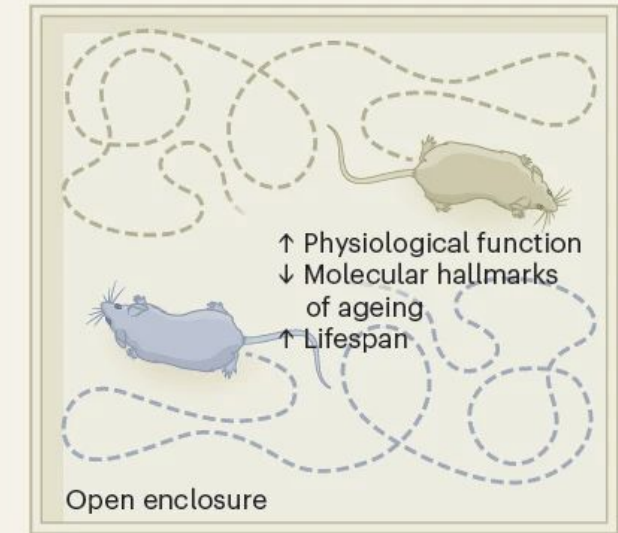
Feng *et al.* put 20-month-old female mice from an inbred laboratory strain on a calorie-restricted diet, or treated them with semaglutide. The results suggest that GLP-1 receptor activation and calorie restriction slow ageing through overlapping but distinct mechanisms.

Y. Feng *et al.* (2026), "Late-life semaglutide treatment slows ageing and extends lifespan in female mice," <https://doi.org/10.1038/s41586-026-10940-7>, Nature; News and Views: [Weight-loss drug slows ageing in female mice](#) by Maria Emilia Fernandez & Rafael de Cabo

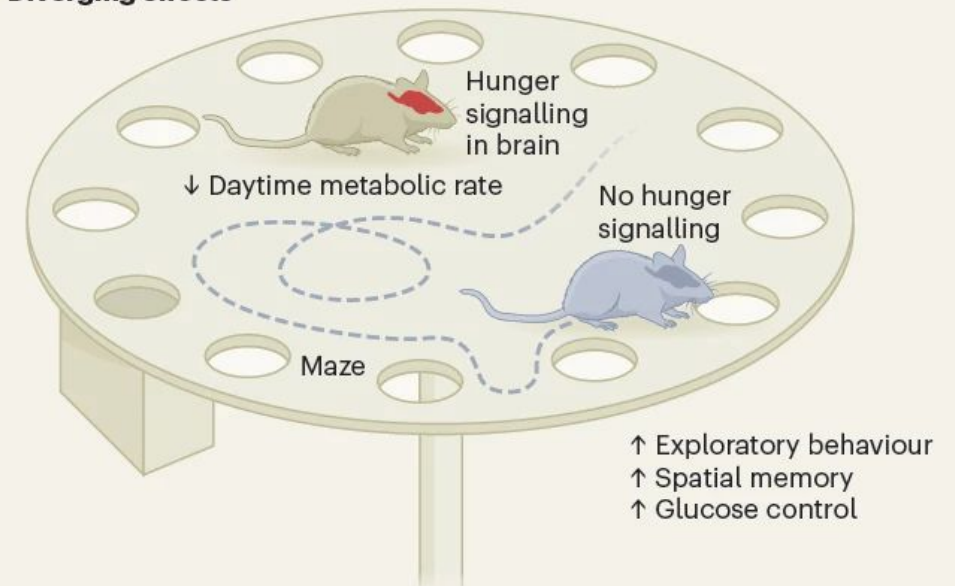
a Condition



b Converging effects



c Diverging effects



GLP1R agonists: biology, alternatives, and social factors

Offline activity:

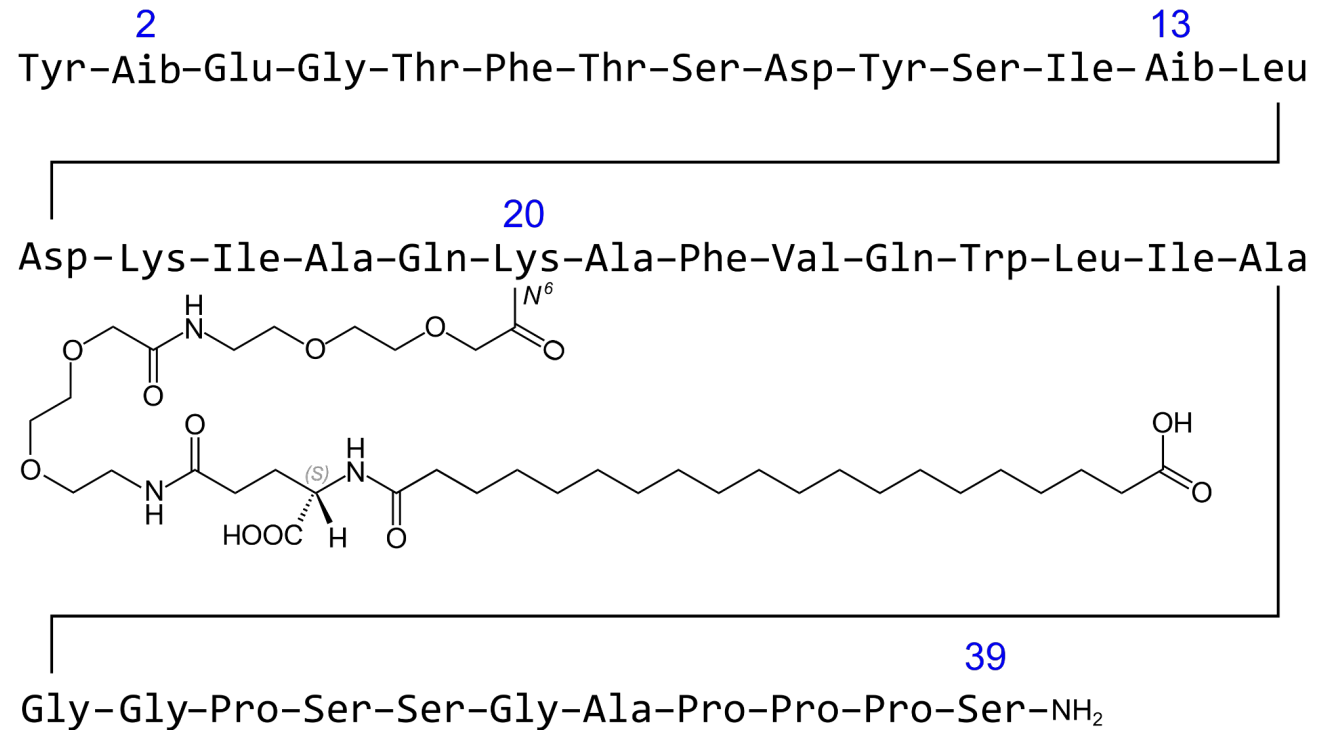
1. Watch the video made by Harvard Medical School Continuing Education (https://www.youtube.com/watch?v=P6gt4A_3Whs).
2. What other possibilities do we have to control weight and improve health? What are the pros and cons compared with pharmacological inventions? What would *you* do?
3. Compare the prevalence of (1) *obesity* and (2) *type 2 diabetes* (T2D) in the world, the US, Switzerland, China, and your country of origin (if not already covered), both now and in the last 30 years. Draw one graph with the data for each condition.
 - a. What pattern do *you* observe?
 - b. What are the factors driving the changes?
 - c. What consequences do the trends have on discovery/development of GLP-1 agonists?
 - d. What consequences does the availability of GLP-1 agonists as drugs have on the trends?

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The structure of Tirzepatide, a dual GIP/GLP-1 receptor agonist (Eli Lilly company)



[Benff, CC BY-SA 4.0](#)